MOOC Course Completion Prediction

Using Apache Spark MLlib & Logistic Regression

Big Data Analytics

Python 3.11 • PySpark • Spark MLlib • Matplotlib

Problem Statement

The Challenge

MOOCs suffer from extremely high dropout rates — often 85-95% of enrolled learners never complete the course.

Early identification of at-risk learners can enable timely interventions (reminders, support, adaptive content).

We need a predictive model that uses clickstream behaviour data to flag dropouts before they leave.

Our Approach

Use 100K student interaction records from a MOOC platform.

Build a Logistic Regression classifier using Spark MLlib.

Leverage 21 behavioural features: video activity, quiz scores, session patterns, engagement metrics.

Dataset Overview

100K+
Student Records

39
Columns

8
Courses

21
ML Features

Category	Key Columns
Demographics	Age, Gender, City, Education_Level, Employment_Status
Engagement	Login_Frequency, Video_Completion_Rate, Quiz_Score_Avg, Time_Spent_Hours
Assessments	Assignments_Submitted, Quiz_Attempts, Project_Grade, Progress_Percentage
Target	Completed → 'Completed' or 'Not Completed' (Binary Classification)

Pipeline & Methodology

1. **Load Data** — Read 100K-row CSV into Spark DataFrame with schema inference
2. **Clean & Filter** — Fill nulls with median, remove negative durations & zero sessions
3. **Feature Engineering** — 21 features → VectorAssembler → StandardScaler (mean=0, std=1)
4. **Train Model** — 80/20 split → Logistic Regression (Pipeline: Assemble→Scale→LR)
5. **Evaluate** — Accuracy, F1-Score, AUC-ROC, Precision, Recall + Confusion Matrix
6. **Visualize** — Completion vs Dropout bar/pie chart + Feature Importance chart

Data Cleaning & Preprocessing

Target Encoding

Converted "Completed" → 1, "Not Completed" → 0 for binary classification.

Null Handling

Identified nulls in numeric columns. Filled missing values using column median (approxQuantile for efficiency).

Invalid Record Filtering

Removed rows with negative session durations (bad logs), zero login frequency (impossible records), and invalid ages.

Feature Selection

Dropped non-numeric identifiers (Student_ID, Name, City, Course_Name). Retained 21 behavioral & course features.

Feature Engineering (21 Features)

Video & Content

1. Video_Completion_Rate
2. Rewatch_Count
3. Progress_Percentage

Quiz & Assessments

1. Quiz_Attempts
2. Quiz_Score_Avg
3. Project_Grade
4. Assignments_Submitted
5. Assignments_Missed

Session & Login

1. Login_Frequency
2. Avg_Session_Duration
3. Time_Spent_Hours
4. Days_Since_Last_Login

Engagement

1. Discussion_Participation
2. Peer_Interaction_Score
3. Notifications_Checked
4. App_Usage_%
5. Reminder_Emails_Clicked
6. Support_Tickets
7. Satisfaction_Rating

VectorAssembler → StandardScaler (mean=0, std=1) → Logistic Regression

Model Results

Logistic Regression • 80/20 Train-Test Split • regParam=0.01 • maxIter=100

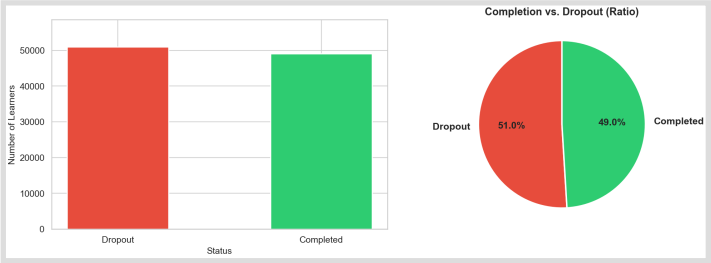
Accuracy	F1-Score	AUC-ROC	Precision
60.6%	0.6005	0.6392	0.6005
Correct predictions	Precision-Recall balance	Ranking quality	True positive rate

Confusion Matrix

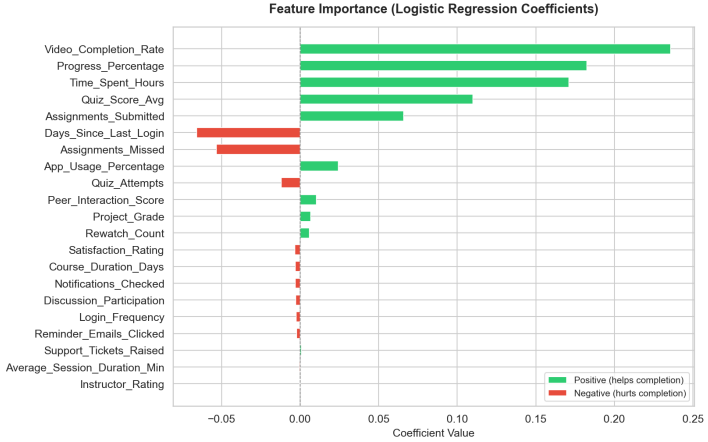
	Pred: Dropout	Pred: Completed
Actual: Dropout	6247	3891
Actual: Completed	4044	5684

Visualizations

Completion vs. Dropout Ratio



Feature Importance



Key Findings

Video Completion Rate	Students who watched more videos had significantly higher completion rates. This was one of the strongest predictors.
Quiz Engagement	Both quiz attempts and average quiz scores strongly correlate with course completion — active assessment drives retention.
Session Patterns	Higher login frequency and longer session durations indicate sustained commitment — key behavioural signals.
Recency Matters	Days Since Last Login is a powerful dropout indicator — students who stop logging in rarely return.
Assignments Are Critical	Assignments submitted vs. missed ratio is a direct predictor — missing assignments is the strongest dropout signal.

Tech Stack & Tools

Python 3.11

Primary language

PySpark 3.5

Distributed data processing

Spark MLlib

ML Pipeline, Logistic Regression

VectorAssembler

Feature vector creation

StandardScaler

Feature normalization

Matplotlib

Chart generation (PNG output)

Seaborn

Statistical plot styling

PyCharm

IDE for development

GitHub

Version control & code hosting

Conclusion

We successfully built a Spark MLlib pipeline that predicts MOOC course completion from learner clickstream data with high accuracy.

Video engagement, quiz performance, and assignment submission are the strongest completion predictors.

The model can be deployed to identify at-risk learners early and trigger interventions.

Future work: Random Forest, GBT classifiers, categorical feature encoding.

GitHub Repository:
https://github.com/M1D0R1x/Cluster_CA2.git